PSEUDO-COHERENCE IS PRESERVED BY DERIVED BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Pseudo-coherence at a degree, tested by a map from a bounded complex of finite free modules with prescribed cohomology behaviour. defined earlier in the same section, source lines 16002-16156. Supplies the witness that is transported through the base change.
- The vanishing statement for the cohomology of the cone of a map that is an isomorphism above a degree. the author cites the dual of Derived Categories, Lemma derived-lemma-negative-vanishing at proof lines 16367-16370. Keeps the acyclicity of the cone after base change.
- Pseudo-coherent complexes are those that are m-pseudo-coherent for all m. the author cites Lemma lemma-pseudo-coherent at proof lines 16378-16380. Derives the unbounded statement from the graded one.

Source reference. Original lemma 0650: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7884–7884. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 064H: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7866–7866. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 05TC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7476–7476. Navigation pointer to a statement the

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counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 064U: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7878–7878. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $A \to B$ be a ring map. Let $K^\bullet$ be an m-pseudo-coherent (resp. pseudo-coherent) complex of A-modules. Then $K^\bullet \otimes^{\mathbf{L}}_A B$ is an m-pseudo-coherent (resp. pseudo-coherent) complex of B-modules.*

Proof. First we note that the statement of the lemma makes sense as $K^\bullet$ is bounded above and hence $K^\bullet \otimes^{\mathbf{L}}_A B$ is defined by Equation (064H). Having said this, choose a bounded complex $E^\bullet$ of finite free A-modules and $\alpha : E^\bullet \to K^\bullet$ with $H^i(\alpha)$ an isomorphism for $i > m$ and surjective for $i = m$. Then the cone $C(\alpha)^\bullet$ is acyclic in degrees $\geq m$. Since $- \otimes^{\mathbf{L}}_A B$ is an exact functor we get a distinguished triangle

$$(E^\bullet \otimes^{\mathbf{L}}_A B, K^\bullet \otimes^{\mathbf{L}}_A B, C(\alpha)^\bullet \otimes^{\mathbf{L}}_A B)$$

of complexes of B-modules. By the dual to Derived Categories, Lemma 05TC we see that $H^i(C(\alpha)^\bullet \otimes^{\mathbf{L}}_A B) = 0$ for $i \geq m$. Since $E^\bullet$ is a complex of projective R-modules we see that $E^\bullet \otimes^{\mathbf{L}}_A B = E^\bullet \otimes_A B$ and hence

$$E^\bullet \otimes_A B \longrightarrow K^\bullet \otimes^{\mathbf{L}}_A B$$

is a morphism of complexes of B-modules that witnesses the fact that $K^\bullet \otimes^{\mathbf{L}}_A B$ is m-pseudo-coherent. The case of pseudo-coherent complexes follows from the case of m-pseudo-coherent complexes via Lemma 064U. □